

Heme Synthesis & Porphyrrias

Darren Browning, PhD • designed to follow alongside the lecture deck

Official Week 7 LOs

Slide-by-slide anchors

Faculty handout details

Board enrichment separated

Factoid sheet companion

Overview: What the Official Learning Objectives Want From You

LO 1. Describe the structure/function of heme and the pathway of heme synthesis.

LO 2. Explain how heme synthesis is regulated in liver and bone marrow.

LO 3. Describe how defects in heme synthesis lead to porphyria, and identify the symptoms of the most common causes: porphyria cutanea tarda, acute intermittent porphyria, erythropoietic protoporphyria, and lead poisoning.

LO 4. Explain the rationale for porphyria treatment, including the use of hemin and glucose.

SCHOLARRX BRICK FOR THIS OVERVIEW

Heme Metabolism - ScholarRx Brick

This is the Brick explicitly paired with this lecture in the official Week 7 objective document.

LO 1

Slides 3-17: heme structure/functions + the 8-enzyme synthesis pathway.

LO 2

Slides 18-19: hepatic ALAS1 vs erythroid ALAS2 regulation.

LO 3

Slides 20-30: AIP, PCT, EPP, and lead toxicity.

LO 4

Slides 21-28: why glucose/hemin suppress AIP attacks and treatment logic for photosensitive porphyrias.

MENTAL MODEL BEFORE YOU START

Build heme -> control the first step -> disease happens when a downstream block makes an upstream intermediate accumulate.

Early soluble precursors (ALA/PBG) tend to cause **neurovisceral** symptoms; later porphyrins tend to cause **photosensitivity**.

Slides 3-7

Heme structure, function, Hb/Mb, and CYP450

LO 1

What the slides are establishing

- **Heme is a porphyrin:** a cyclic tetrapyrrole made from four pyrrole rings linked by methine bridges, with a central metal-binding core.
- Heme functions as a **prosthetic group** that lets proteins bind oxygen and/or transfer electrons.
- The faculty list includes cytochrome c (ETC), guanylate cyclase (cGMP), catalase (H₂O₂ degradation), tryptophan pyrrolase, CYP450, hemoglobin, and myoglobin.
- Hemoglobin transports oxygen in blood; myoglobin stores/buffers oxygen in muscle.
- Hepatic CYP450 proteins perform phase I metabolism and are induced by xenobiotic-sensing nuclear receptors such as PXR/CAR.

WHAT THIS MEANS

Heme is useful because the iron can participate in reversible oxygen binding and redox chemistry. That is why one small cofactor appears in proteins doing very different jobs: oxygen delivery, electron transport, peroxide handling, and drug metabolism.

DO NOT MISS FROM THE FACULTY HANDOUT

- Heme side chains are not all equivalent: methyl/vinyl groups are relatively hydrophobic, while propionate groups are hydrophilic, helping orient heme inside an apoprotein pocket.
- Hemoglobin/myoglobin bind oxygen to **Fe²⁺**. Oxidation to **Fe³⁺** produces methemoglobin, which cannot carry oxygen until reduced back to Fe²⁺.
- Faculty handout: CYP450 enzymes are abundant in liver ER/mitochondria and inducible by lipophilic xenobiotics.

BOARD/CLINICAL ENRICHMENT

Think “heme = iron chemistry.” Fe²⁺ is the oxygen-binding state in Hb/Mb; Fe³⁺ is methemoglobin. This is related to heme biology but not a major porphyria focus of this lecture.

LO 1 checkpoint - structure/function

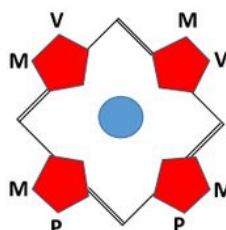
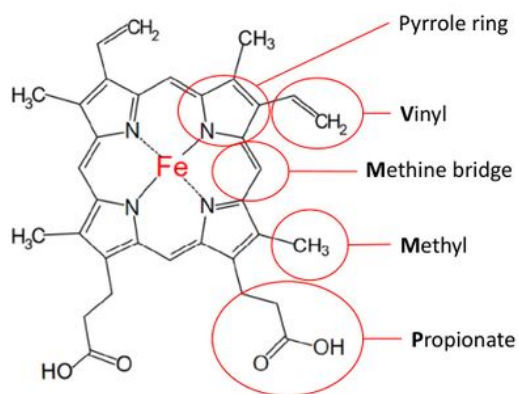
Can you explain why a porphyrin + iron can support oxygen binding/electron transfer, and name at least four major heme-containing proteins with their functions?

Slides 8-17

Build heme: the complete pathway

LO 1

How to build heme in 6 easy steps



- 1- Make the building blocks
- 2- Make the pyrrole ring
- 3- Link 4 pyrroles together then cyclize them to make the tetrapyrrole
- 4- Modify the chains M,P, V
- 5- Form Methine bridges
- 6- Add iron to the core

Slide 8. Use this as the visual skeleton: make ALA -> make a pyrrole -> join four pyrroles -> modify side chains -> create the conjugated ring -> insert iron.

NEED TO MEMORIZE COLD

Location pattern: mitochondria starts it, cytosol builds the middle, mitochondria finishes it. Faculty handout specifies **step 1 and the final 3 steps are mitochondrial**; steps 2-5 are cytosolic.

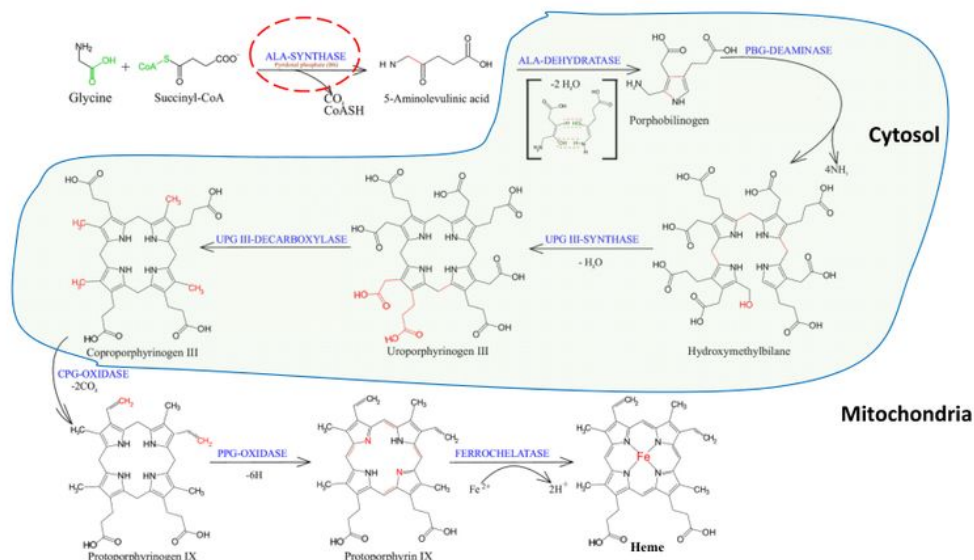
#	Enzyme	Reaction / product	Location	High-yield hook
1	ALA synthase (ALAS)	Glycine + succinyl-CoA -> delta-ALA	Mito	Rate-limiting. Requires pyridoxal phosphate (B6). ALAS1 liver; ALAS2 erythroid.
2	ALA dehydratase (ALAD)	2 ALA -> porphobilinogen (PBG)	Cytosol	Zinc-dependent; lead inhibits.

3	Porphobilinogen deaminase (PBGD/ HMBS)	4 PBG -> hydroxymethylbilane	Cytosol	AIP when deficient.
4	Uroporphyrinogen III synthase	Hydroxymethylbilane -> uroporphyrinogen III	Cytosol	Cyclizes the linear tetrapyrrole.
5	Uroporphyrinogen decarboxylase (UROD)	Uroporphyrinogen III -> coproporphyrinogen III	Cytosol	PCT when activity reduced.
6	Coproporphyrinogen oxidase	Coproporphyrinogen III -> protoporphyrinogen IX	Mito	Converts propionate side chains to vinyl groups.
7	Protoporphyrinogen oxidase	Protoporphyrinogen IX -> protoporphyrin IX	Mito	Creates the conjugated porphyrin system / methine bridges.
8	Ferrochelatase	Protoporphyrin IX + Fe ²⁺ -> heme	Mito	EPP when deficient; lead inhibits .

FACULTY HANDOUT DETAIL THAT IS EASY TO MISS

- **Porphyrinogens are colorless.** Oxidation can generate reddish porphyrins.
- Hydroxymethylbilane can spontaneously cyclize into a nonproductive uroporphyrinogen I isomer; the enzyme-directed product for heme synthesis is uroporphyrinogen III.
- When iron is unavailable, ferrochelatase can insert zinc, forming **zinc protoporphyrin**.

Mitochondrial ALAS is the rate limiting step controlling heme synthesis



Slide 17. Faculty's integrated pathway. This is worth returning to after learning the disease blocks because the defects map directly onto it.

COMMON EXAM TRAP

Do not reduce the pathway to only "ALAS → ALAD → ferrochelatase." The official LO says *describe the pathway*, so know the major intermediates and the three disease-defining enzymes: **PBG deaminase**, **UROD**, **ferrochelatase**, plus both lead targets.

LO 1 checkpoint - pathway

Without looking, can you draw a line from glycine + succinyl-CoA to heme, place the pathway in mitochondria/cytosol, and mark AIP, PCT, EPP, and lead on it?

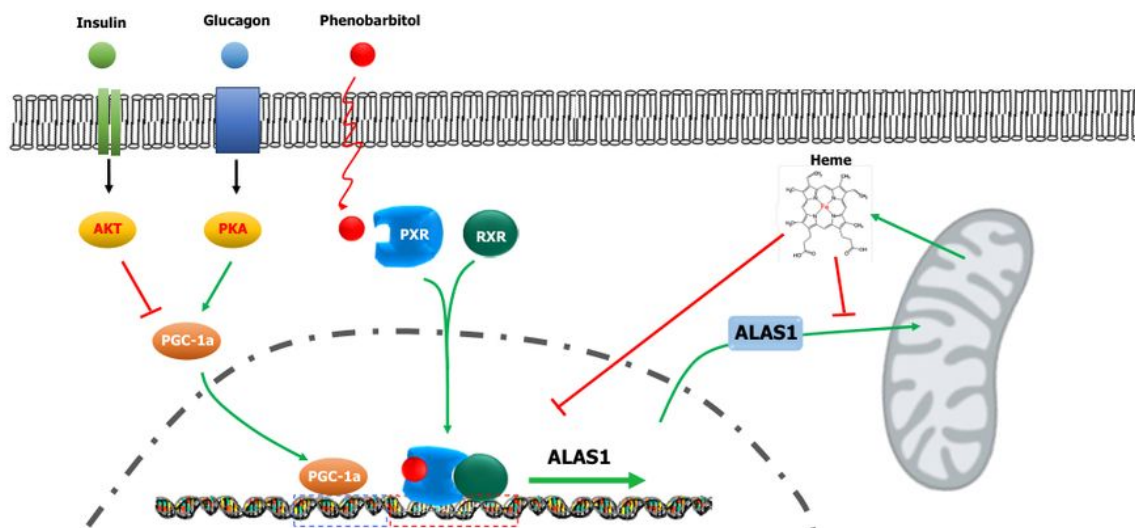
ALAS1 - liver

Main purpose: supply heme for CYP450 and other hepatic hemoproteins.
Controlled strongly by the **regulatory heme pool** and metabolic/xenobiotic signals.

ALAS2 - erythroid

Main purpose: coordinate heme production with hemoglobin synthesis.
Controlled by **iron availability** and **EPO/JAK-STAT**.

Regulation of hepatic ALAS1



Slide 18. Hepatic ALAS1: heme feedback inhibits; fasting/glucagon-PGC-1alpha and xenobiotic receptor signaling increase expression; insulin opposes the fasting program.

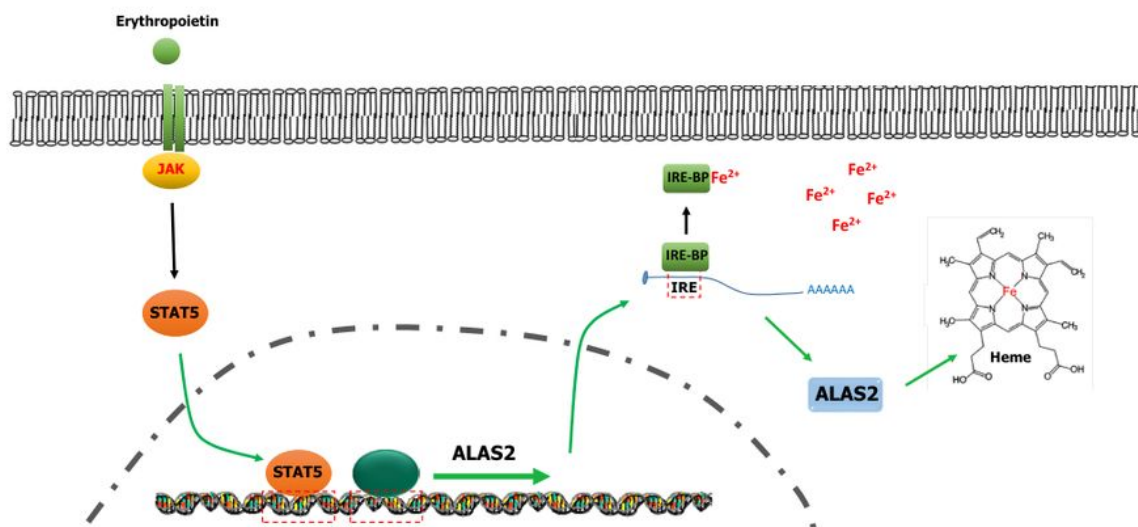
Hepatic ALAS1 - causal chain

Fasting -> glucagon/PKA -> PGC-1alpha -> increased ALAS1. CYP450 inducers such as phenobarbital activate xenobiotic receptors (PXR/CAR) and increase ALAS1. Heme feeds back to suppress ALAS1.

FACULTY HANDOUT: HEME FEEDBACK IS MULTI-LEVEL

Excess heme can directly inhibit ALAS1, reduce transcription, reduce mRNA stability, block mitochondrial import of the precursor, and promote proteolysis. You do not necessarily need all five mechanisms to answer the LO, but understand that feedback is **strong and redundant**.

Regulation of Erythropoietic ALAS2



Slide 19. Erythroid ALAS2: EPO drives JAK/STAT5-dependent transcription; iron controls translation through an IRE/iron-regulatory protein mechanism.

NEED TO MEMORIZE - ALAS1 VS ALAS2

ALAS1 = liver = heme feedback + fasting/drugs.

ALAS2 = erythroid = EPO + iron availability.

WHY THIS MATTERS CLINICALLY

This regulation explains both **AIP triggers** and **AIP treatment**. Anything that drives hepatic ALAS1 pushes more precursor into a blocked pathway; hemin and carbohydrate loading push ALAS1 back down.

LO 2 checkpoint

Can you compare ALAS1 and ALAS2 in one sentence each and explain why fasting, phenobarbital, glucose, hemin, iron, and EPO move the pathway in different directions?

Slides 20-22

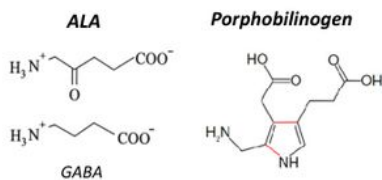
Porphyria logic + Acute Intermittent Porphyria

LO 3 + LO 4

PATTERN RECOGNITION

AIP = neurovisceral, no photosensitivity. Think severe abdominal pain + autonomic/neurologic/psychiatric symptoms + precipitating ALAS1 trigger.

Acute Intermittent Porphyria (AIP)



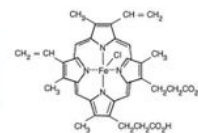
Neuropsychiatric Symptoms:

- Initiated by lipophilic drugs, steroid hormones, fasting
- Abdominal pain, tachycardia
- confusion, paranoia, psychosis
- peripheral neuropathy, proximal limb paresis
- High ALA & PBG levels in urine.
- Reddish-brown colored urine after sunlight exposure (porphobilinogen is oxidized).



Treatment:

- Avoid aggravating factors
- Infusion of glucose (400g/day, 1-2 days)
- Intravenous infusion of Hemin (1-4mg/kg/day, 3-14 days)
- Liver transplant



Slide 21. AIP: partial PBG deaminase deficiency -> ALA/PBG accumulation -> neuropsychiatric/neurovisceral symptoms. The slide also shows the treatment logic.

Feature	AIP
Defect	Partial deficiency of porphobilinogen deaminase (PBGD; hydroxymethylbilane synthase/HMBS), classically autosomal dominant
Accumulation	ALA + PBG
Target syndrome	Acute neurovisceral attacks
Triggers	Fasting; CYP450-inducing lipophilic drugs; steroid hormones; faculty examples include phenobarbital
Clues	Severe abdominal pain, tachycardia, nausea/vomiting/constipation, confusion/paranoia/psychosis, neuropathy, proximal paresis, seizures
First-line diagnostic clue	Elevated urinary PBG during attack; urine can darken reddish-brown after light exposure
Treatment	Remove trigger; carbohydrate/glucose; IV hemin; liver transplant in severe recurrent refractory disease

LO 4 - WHY GLUCOSE WORKS

Fasting/glucagon activates PGC-1alpha and ALAS1. Providing metabolizable carbohydrate suppresses the fasting program and **reduces hepatic ALAS1 drive**, so fewer toxic precursors enter the blocked pathway.

LO 4 - WHY HEMIN WORKS

Hemin supplies a heme signal that restores **negative feedback on hepatic ALAS1**. Less ALAS1 activity -> less ALA/PBG production upstream of the PBGD block.

PROFESSOR-SPECIFIC NUMERIC DETAIL

Slide 21 lists glucose **400 g/day for 1-2 days** and hemin **1-4 mg/kg/day for 3-14 days**. Know the treatment rationale first; memorize exact dosing only if your course has emphasized dosing details.

FACULTY HANDOUT NUANCE

AIP generally lacks anemia because the disease-causing mutation typically affects the hepatic PBGD isoform rather than the erythroid isoform. The handout also notes ALA can interfere with GABA signaling, one proposed contributor to neurologic symptoms.

LO 3/4 checkpoint - AIP

Can you explain the entire vignette chain: trigger -> ALAS1 increases -> PBGD block -> ALA/PBG accumulate -> abdominal/neuropsychiatric syndrome -> urine PBG -> glucose/hemin suppress ALAS1?

Slides 23-24

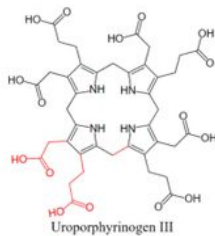
Porphyria Cutanea Tarda (PCT)

LO 3

PATTERN RECOGNITION

PCT = delayed blistering photosensitivity in adults. Think fragile sun-exposed skin, vesicles/erosions/scarring, pigment change, hypertrichosis.

Porphyria cutanea tarda (PCT)



UPG
decarboxylase

Symptoms:

- Delayed photosensitivity (adult onset)
- Skin fragility, blistering, ulcers, scarring
- Hyper and hypopigmentation
- hypertrichosis (periorbital/temporal)
- Urine polycarboxylate porphyrins



Slide 24. PCT clinical phenotype and UROD block. The faculty images highlight blistering/fragility rather than the immediate burning phenotype of EPP.

Feature	PCT
Defect	Reduced hepatic uroporphyrinogen decarboxylase (UROD) activity
Epidemiology	Most common porphyria. Faculty handout: mostly acquired/sporadic; minority inherited AD
Accumulation	Uroporphyrinogen -> oxidized uroporphyrins; elevated urinary polycarboxylate porphyrins
Clinical	Delayed blistering photosensitivity, skin fragility, ulcers/erosions, slow healing/scarring, hyper/hypopigmentation, hypertrichosis
Associations/triggers	Alcohol, estrogen, lipophilic chemicals; strong faculty-handout associations with hepatitis C and iron overload
Management	Remove triggers, limit light exposure, phlebotomy to reduce iron, low-dose chloroquine

DO NOT MISS - UNUSUAL SUNSCREEN DETAIL FROM FACULTY HANDOUT

PCT photosensitization is driven by porphyrins absorbing **visible light** (faculty handout notes an absorption peak around 405 nm). Standard sunscreen is designed mainly for UV and may not fully protect; the handout specifically recommends opaque physical blockers such as zinc/titanium oxide and protective clothing.

LO 3 checkpoint - PCT

Can you distinguish PCT from EPP using **age, timing after light exposure, blistering/scarring, accumulated metabolite, diagnostic specimen, and enzyme defect**?

Slides 25-28

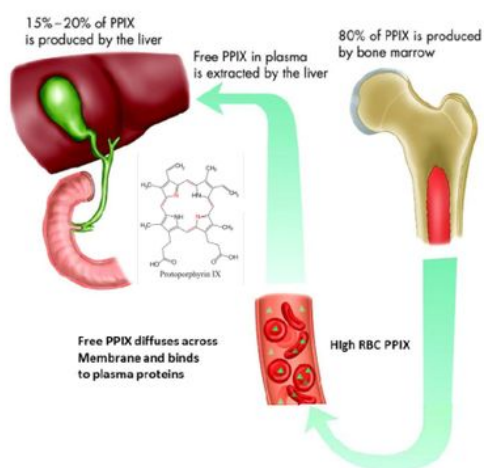
Erythropoietic Protoporphyria (EPP)

LO 3

PATTERN RECOGNITION

EPP = immediate painful non-blistering photosensitivity, often beginning in childhood.

Erythropoietic protoporphyria (EPP): Ferrochetalase deficiency



Symptoms:

- Acute non-blistering photosensitivity (pediatric onset).
- Swelling, burning, itching, redness of the skin.
- Subside in 1-2 days without scarring.
- Excess protoporphyrin in the blood plasma and RBC
- Reduced liver function due to accumulation of insoluble protoporphyrin IX in hepatocytes.
- Protoporphyrin is excreted by the liver into bile/feces, possible gallstones.

Slide 26. Most protoporphyrin IX is produced by bone marrow, circulates in RBC/plasma, is taken up by liver, and is eliminated through bile/feces.

Feature	EPP
Defect	Reduced ferrochelatase activity
Accumulation	Protoporphyrin IX in bone marrow, RBCs, plasma, and potentially liver
Clinical	Burning/stinging, itching, redness, swelling soon after visible-light exposure; no blisters ; typically resolves without scarring
Age	Often infancy/childhood; faculty handout calls it the most common porphyria in children
Diagnosis	Total blood/RBC protoporphyrin; faculty notes liver biopsy can show rust-brown deposits and birefringent Maltese-cross material in hepatic disease

Why urine is not the main specimen	Protoporphyrin IX is hydrophobic/water-insoluble and is excreted mainly in bile/feces
Complications	Protoporphyrin gallstones, cholestasis, progressive liver disease in some patients
Treatment	Light avoidance; beta-carotene; afamelanotide (MC1-R agonist) to increase melanin/pain-free light exposure

PCT VS EPP - FASTEST DISTINCTION

PCT: delayed + blistering/scarring + adult. **EPP:** immediate burning + non-blistering + childhood.

FACULTY HANDOUT DETAIL

EPP protoporphyrin is hydrophobic. The handout uses this to explain both the tissue distribution and why EPP differs from water-soluble porphyrin disorders in urinary excretion and skin injury pattern.

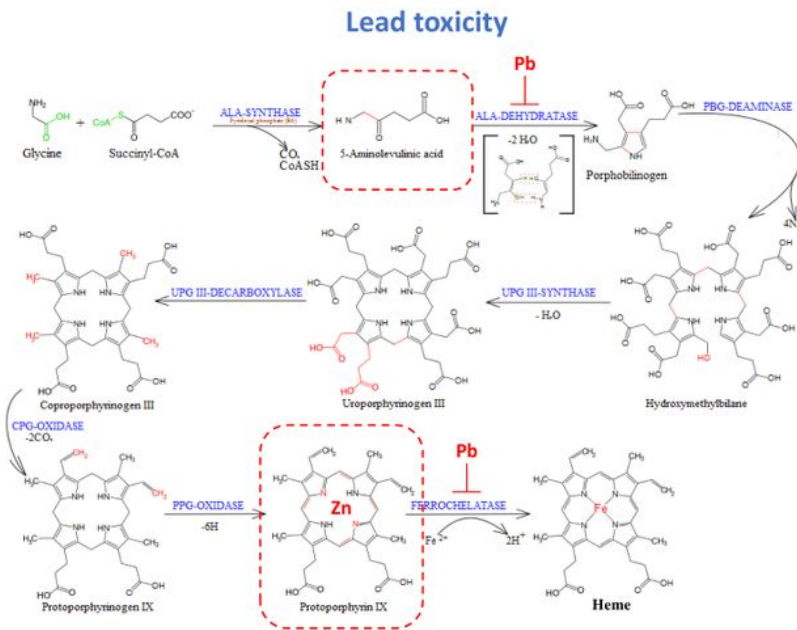
LO 3 checkpoint - EPP

If a child has burning and swelling within minutes of sunlight but no blisters, can you identify ferrochelatase, protoporphyrin IX, blood testing, and possible hepatobiliary complications?

Slides 29-30

Lead toxicity as a heme-synthesis disorder

LO 3



Slide 29. Lead hits the pathway twice: ALA dehydratase and ferrochelatase.

NEED TO MEMORIZE COLD

Lead inhibits ALA dehydratase + ferrochelatase. Therefore: urinary ALA rises and erythrocyte/zinc protoporphyrin rises.

Faculty slide symptoms

Abdominal pain, headache, numbness/tingling, weakness/fatigue, memory loss and mood/irritability changes. The slide emphasizes increased urinary delta-ALA and increased erythrocyte protoporphyrins.

FACULTY HANDOUT MECHANISM

- Lead can displace the zinc required by ALA dehydratase.
- At ferrochelatase, lead favors formation/accumulation of **zinc protoporphyrin** rather than normal heme.
- Blood lead level is the definitive measurement; zinc protoporphyrin reflects chronic effect on heme synthesis.
- Heme inhibition produces **microcytic, hypochromic anemia**.

BOARD ENRICHMENT FROM THE COMPANION STUDY GUIDE

Classic board stems may additionally show **basophilic stippling**, sideroblastic/ring-sideroblast changes, and motor neuropathy such as wrist/foot drop. These are useful recognition clues but are not all explicitly pictured on Browning's main slides.

COMMON TRAP

Lead inhibits ferrochelatase like EPP, but lead poisoning is **not the same phenotype as EPP**. Faculty handout explains that zinc protoporphyrin is less phototoxic and remains associated with erythrocytes, so lead does not produce the classic immediate EPP photosensitivity syndrome.

LO 3 checkpoint - lead

Can you identify both inhibited enzymes, predict which intermediates rise, and connect heme failure to a microcytic/hypochromic anemia plus neuro/GI symptoms?

Learning-Objective Synthesis

Official LO	What you should be able to do without notes
LO 1	Define heme/porphyrin, name major hemoprotein functions, draw the 8-enzyme pathway with cellular locations, and identify ALAS as rate-limiting.
LO 2	Contrast hepatic ALAS1 (heme feedback, fasting/glucagon/PGC-1alpha, CYP450-inducing drugs) with erythroid ALAS2 (iron/IRE + EPO/JAK-STAT).

LO 3	For AIP, PCT, EPP, and lead: state the defective enzyme(s), accumulated metabolites, clinical pattern, best first-line diagnostic clue, and major management approach.
LO 4	Explain mechanistically why glucose and hemin decrease precursor production in AIP by suppressing hepatic ALAS1.

One-table disease comparison

	AIP	PCT	EPP	Lead
Defect	PBG deaminase/ HMBS	UROD activity	Ferrochelatase	ALAD + ferrochelatase inhibited
Build-up	ALA + PBG	Uroporphyrins	Protoporphyrin IX	ALA + zinc/erythrocyte protoporphyrin
Signature	Neurovisceral, no photosensitivity	Delayed blistering photosensitivity	Immediate painful non-blistering photosensitivity	GI/neuro + microcytic hypochromic anemia
Test clue	Urine PBG	Urine uroporphyrin	Blood/RBC protoporphyrin	Blood lead; zinc protoporphyrin/ALA supportive
Treatment logic	Remove trigger, glucose, hemin	Remove triggers, light protection, phlebotomy/low-dose chloroquine	Light protection, beta-carotene, afamelanotide	Remove exposure; treatment depends on toxicity severity

LAST 60 SECONDS

AIP = PBG deaminase = abdominal/psych/neuro = urine PBG = glucose/hemin.

PCT = UROD = adult blistering photosensitivity = urine uroporphyrin.

EPP = ferrochelatase = child immediate burning/no blisters = RBC protoporphyrin.

Lead = ALAD + ferrochelatase = ALA + zinc protoporphyrin + microcytic anemia.